

INDU 211 · Comprehensive Topic Guide (Part 6)

Chapters 14 & 15: Linear Programming & Queuing Models

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Built on the teacher's Lecture 9.0 (Chapter 14, Deterministic Operations Research) and Lecture 10 (Chapter 15, Probabilistic Models: Queuing Theory), with depth from the course textbook (Hicks, Chapters 14 and 15).

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1. What Operations Research Is

Operations research (OR) is the scientific study of operations, which started during World War II with the allocation of scarce military resources. It builds a **mathematical model** that captures the essence of a real problem and solves it for the best decision (Lecture 9.0, slide 3).

Deterministic OR (Chapter 14)	Probabilistic OR (Chapter 15)
All data known with certainty	Some data are random (arrivals, service times, demand)
Linear, integer, nonlinear, dynamic programming; transportation, assignment, shortest path, TSP/VRP	Queuing theory, inventory under uncertainty, Markov chains, simulation

From the textbook (Hicks, §14.1): the UK Operational Research Society defines OR as "the attack of modern science on complex problems arising in the direction and management of large systems of men, machines, materials and money." Its tools overlap heavily with industrial engineering, which is why IEs use them daily.

Applications (slide 4): supply chain, production and inventory planning, facility location and layout, transportation and logistics, personnel scheduling, timetabling, job scheduling, energy management, robotics, product design.

2. Anatomy of a Linear Program

Every mathematical programming model has **decision variables, parameters, an objective function and constraints** (slide 5). It is a **linear** program when the objective and every constraint are linear in the decision variables.

$$\text{Maximise (or minimise) } Z = C_1X_1 + C_2X_2 + \cdots + C_nX_n$$

$$A_{11}X_1 + A_{12}X_2 + \cdots + A_{1n}X_n \leq B_1$$

$$\begin{array}{l} \text{subject to} \\ A_{m1}X_1 + A_{m2}X_2 + \cdots + A_{mn}X_n = B_m \\ X_1, X_2, \dots, X_n \geq 0 \end{array}$$

Symbol	Meaning
X_j	Decision variables (e.g. units of product j)
C_j	Contribution of one unit of X_j to profit or cost
A_{ij}	Amount of resource i used by one unit of X_j
B_i	Amount of resource i available
≥ 0	Non-negativity: you cannot produce negative units

Constraints can be $\leq$, $\geq$ or $=$, in any combination.

3. Worked Example: Lawn Grow Product Mix

Problem (slide 9): Lawn Grow makes two riding mowers, the **Max** (profit \$400) and the **Multimax** (profit \$800). Each month there are 5,000 hours of fabrication and 3,000 hours of assembly. How many of each should be made to maximise profit?

Constraints

Given the resource information below from the problem:

Max (X1)	Multimax (X2)		
Required Time/Unit	Required Time/Unit	Available Time/Month	
3	5	5,000	Fab
1	4	3,000	Assy

We can now state the constraints and non-negativity requirement:

$$3X_1 + 5X_2 \leq 5,000 \quad (\text{Fab.})$$

$$X_1 + 4X_2 \leq 3,000 \quad (\text{Assy.})$$

$$X_1, X_2 \geq 0 \quad (\text{Non-negativity})$$

Note that the inequalities are less-than-or-equal since the time

Figure 1: Constraints, Lecture 9.0, slide 11.

Reading the table: each **row** is a resource and becomes one constraint; each **column** is a product and becomes one variable. Fabrication: 3 h per Max and 5 h per Multimax, 5,000 h available. Assembly: 1 h and 4 h, 3,000 h available.

Step 1: variables. X_1 = Max per month, X_2 = Multimax per month.

Step 2: objective.

$$\max Z = 400X_1 + 800X_2$$

Step 3: constraints.

$$3X_1 + 5X_2 \leq 5000 \quad (\text{fabrication})$$

$$X_1 + 4X_2 \leq 3000 \quad (\text{assembly})$$

$$X_1, X_2 \geq 0$$

The inequalities are $\leq$ because hours used cannot exceed the hours available.

4. The Graphical Method, Step by Step

For two variables the LP can be solved on a graph (slide 15):

1. Draw each constraint as a line by finding its intercepts: $3X_1 + 5X_2 = 5000$ meets the axes at (1666.7, 0) and (0, 1000); $X_1 + 4X_2 = 3000$ at (3000, 0) and (0, 750).
2. Shade the side that satisfies $\leq$ for both lines: that common area is the **feasible region**.

3. Draw an objective line for any value, say $400X_1 + 800X_2 = 8000$, to see its direction.
4. Slide it parallel in the direction of increasing profit until it last touches the feasible region.
5. That **corner point** is the optimum.

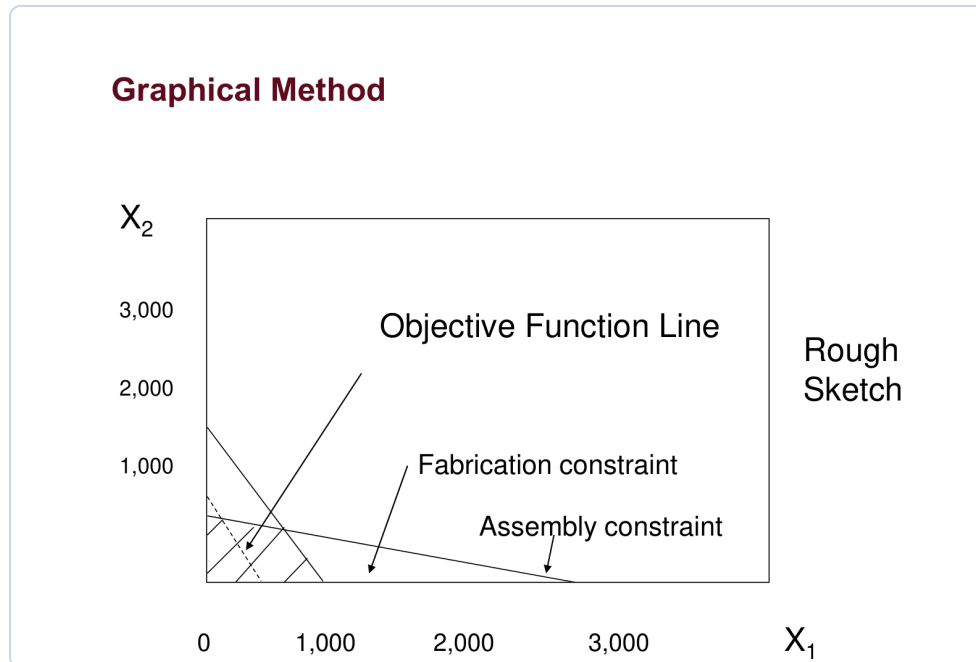


Figure 2: Graphical method, Lecture 9.0, slide 16.

Reading the sketch: the fabrication line is steeper, the assembly line is flatter, and the feasible region is the area under both, next to the origin. The objective line is pushed up and to the right until it touches the corner where the two constraints cross.

Solving the corner (slide 17):

$$X_1 = 3000 - 4X_2 \Rightarrow 3(3000 - 4X_2) + 5X_2 = 5000 \Rightarrow 7X_2 = 4000 \Rightarrow X_2 = 571.4, X_1 = 714.3$$

$$Z = 400(714.3) + 800(571.4) \approx \$742,857$$

Checking the other corners confirms it: $(0, 750)$ gives \$600,000 and $(1666.7, 0)$ gives \$666,667. (The slide rounds to 715 and 571 for \$742,800; strictly, 715 uses 5,002 fabrication hours, so 714 is the feasible whole number.)

Why the optimum is always at a corner: the objective is a straight line and the region is bounded by straight lines, so the last point the sliding line touches is a vertex (or a whole edge, if the objective is parallel to it, giving multiple optima).

5. Minimisation and Special Cases

Lecture Example 2 (slides 18–19):

$$\min Z = 5x_1 + 4x_2 \quad \text{s.t.} \quad 5x_1 + x_2 \geq 5, \quad x_1 + 5x_2 \geq 5, \quad x_1, x_2 \geq 0$$

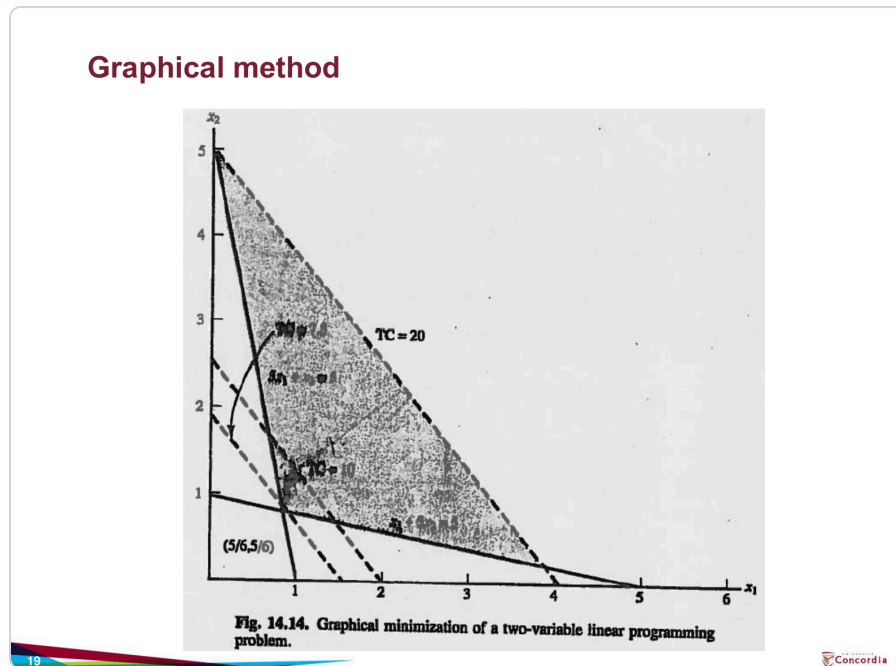


Figure 3: Graphical minimisation of a two-variable LP, Lecture 9.0, slide 19 (textbook Fig. 14.14).

Reading the graph: with $\geq$ constraints the feasible region lies **above** both lines (shaded) and is unbounded. The cost lines $TC = 10$ and $TC = 20$ (dashed) are moved **toward the origin**, and the last feasible point they touch is the corner $(5/6, 5/6)$:

$$5x_1 + x_2 = 5, \quad x_1 + 5x_2 = 5 \Rightarrow x_1 = x_2 = \frac{5}{6}, \quad Z = 5\left(\frac{5}{6}\right) + 4\left(\frac{5}{6}\right) = 7.5$$

Equality constraints (Fall 2020 final): a constraint like $x_1 + 3x_2 = 9$ restricts the feasible set to a **line segment**, so the optimum is at one of the segment's endpoints.

Other deterministic models (slide 20, textbook §14.8–14.9): transportation and assignment problems (special LPs), shortest path, minimum spanning tree, TSP/VRP, integer and mixed-integer programming (whole-number or yes/no decisions), nonlinear and dynamic programming. Real problems are solved with software such as CPLEX, Gurobi, MATLAB or Python (slide 21).

6. Queuing Systems: Structure and Randomness

Queuing theory is the mathematical analysis of waiting lines, in emergency rooms, banks, call centres, networks and production lines (Lecture 10, slide 3). Its goal is to minimise the **sum of customer waiting cost and service capacity cost**: more servers mean shorter waits but higher cost.

Elements of Queuing Systems

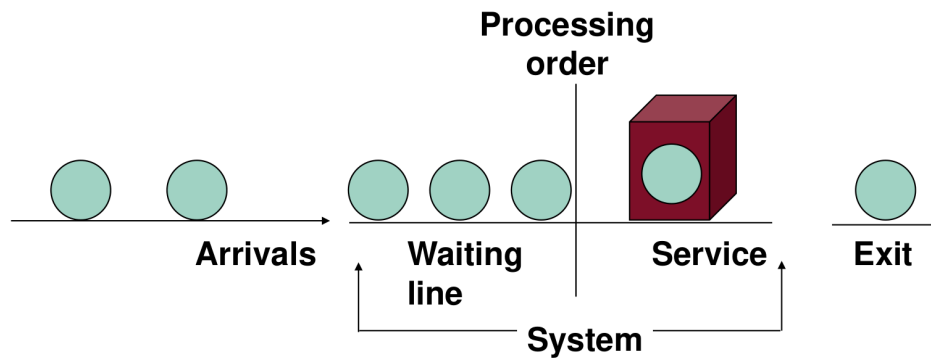


Figure 4: Elements of queuing systems, Lecture 10, slide 5.

Reading the figure: customers **arrive**, wait in the **waiting line** in some processing order, receive **service**, and **exit**. The "system" is the waiting line plus the service.

System characteristics (slide 4): population source (infinite or finite), number of servers (channels), arrival and service patterns, and queue discipline (e.g. first come, first served).

Examples of Queue Structures

	Single Phase	Multiphase
Single Channel	One-person barber shop	Car wash
Multi-channel	Bank tellers' windows	Hospital admissions

Figure 5: Examples of queue structures, Lecture 10, slide 7.

	Single phase	Multiphase
Single channel	One-person barber shop	Car wash
Multi-channel	Bank tellers' windows	Hospital admissions

Modelling randomness

- **Arrivals:** the number of arrivals per unit time follows a **Poisson** distribution with mean rate λ :

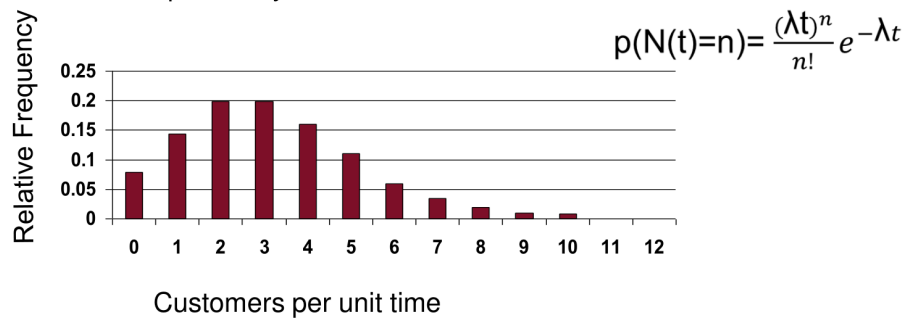
$$P(N(t) = n) = \frac{(\lambda t)^n}{n!} e^{-\lambda t}$$

- **Service times:** follow an **exponential** distribution with mean $1/\mu$ (service rate μ):

$$f(t) = \mu e^{-\mu t}$$

Arrival Rate: Poisson Distribution

- Discrete probability distribution
 - The probability of a given number of events occurring in a fixed interval of time at a known constant mean rate (λ), independently of the time since the last event



- Exponential distribution models the time between these events

Figure 6: Arrival rate, Poisson distribution, Lecture 10, slide 9.

Service Time: Exponential Distribution

- Continuous probability distribution
 - Models the time elapsed between events in a Poisson process
 - Average processing time ($1/\mu$) $\rightarrow$ Average service rate (μ)

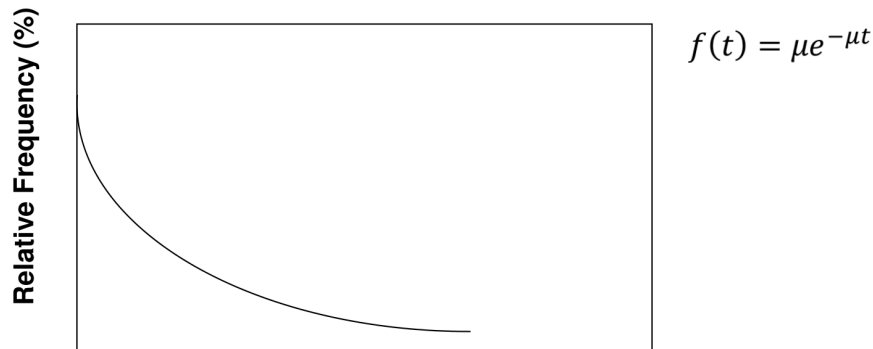


Figure 7: Service time, exponential distribution, Lecture 10, slide 10.

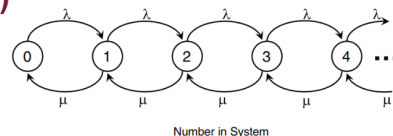
Reading the two plots: the Poisson bars show how many customers arrive per period (most often around the mean, sometimes more). The exponential curve shows that most services are short and a few are long. The two are linked: if arrivals per period are Poisson, the times *between* arrivals are exponential.

7. The M/M/1 Queue

"M/M/1" means Markovian (Poisson) arrivals, Markovian (exponential) service, 1 server.

Model: Single-Service Channel, exponential distribution of arrival and service times (M/M/1)

- Markov chains



- Probability of no customer in the system

$$P(0) = 1 - \lambda / \mu = 1 - \rho$$

- Probability of exactly n customers in the system

$$P(n) = (1 - \rho) \rho^n$$

- The expected number of customers in the system

$$N_s = \rho / (1 - \rho) = \lambda / (\mu - \lambda)$$

- Other service measures can be obtained from the basic relationship

Figure 8: Single-service channel model (M/M/1), Lecture 10, slide 14.

Reading the chain: each circle is a state (the number of customers in the system). Arrivals move the system one state right at rate λ ; service completions move it one state left at rate μ . At steady state, the flow into each state equals the flow out, which gives the geometric distribution below.

Measure	Formula
Utilisation (must be < 1)	$\rho = \frac{\lambda}{\mu}$
Probability the system is empty	$P(0) = 1 - \rho$
Probability of exactly n in the system	$P(n) = (1 - \rho) \rho^n$
Average number in the system	$N_s = \frac{\lambda}{\mu - \lambda}$
Average number in the queue	$N_q = \frac{\lambda^2}{\mu(\mu - \lambda)}$
Average time in the system	$T_s = \frac{1}{\mu - \lambda}$
Average time in the queue	$T_q = \frac{\lambda}{\mu(\mu - \lambda)}$

Steady State Results for Single Service Channel System with exponential (Poisson) arrival and exponential service time

- Utilization (should be < 1): $\rho = \lambda / \mu$
- Average Number in Line (N_q): $N_q = \lambda^2 / \mu (\mu - \lambda)$
- Average Number in System: $N_s = \lambda / (\mu - \lambda)$
- Average Time for Customers:
 - ➔ Wait in Line: $T_q = \lambda / \mu (\mu - \lambda)$
 - ➔ Wait in System: $T_s = 1 / (\mu - \lambda)$

Figure 9: Steady-state results for M/M/1, Lecture 10, slide 16.

Two relationships tie them together:

- **Little's law:** $N_s = \lambda T_s$ and $N_q = \lambda T_q$.
- **Service time:** $T_s = T_q + \frac{1}{\mu}$ (time in the system = wait + one service).

From the textbook (Hicks, §15.2.2): these two relationships hold for all the standard models, "thus, if we calculate the expected number of units, it is easy to get expected time, or vice versa". Compute one measure, then derive the rest.

8. Worked Example: Drive-Up Window

Problem (slide 17): customers arrive at 25 per hour; one employee serves a customer every 2 minutes; Poisson arrivals and exponential service.

Step 1: rates in the same unit.

$$\lambda = 25 \text{ /h}, \quad \mu = \frac{60 \text{ min/h}}{2 \text{ min}} = 30 \text{ /h}$$

Step 2: utilisation.

$$\rho = \frac{25}{30} = 0.833$$

Step 3: numbers in line and in system.

$$N_q = \frac{25^2}{30(30 - 25)} = \frac{625}{150} = 4.17, \quad N_s = \frac{25}{30 - 25} = 5$$

Step 4: times.

$$T_q = \frac{25}{30(5)} = 0.1667 \text{ h} = 10 \text{ min}, \quad T_s = \frac{1}{5} = 0.2 \text{ h} = 12 \text{ min}$$

Check: $T_s - T_q = 2 \text{ min} = 1/\mu \checkmark$, and $N_s = \lambda T_s = 25 \times 0.2 = 5 \checkmark$.

Step 5: probabilities.

$$P(2) = \left(1 - \frac{25}{30}\right) \left(\frac{25}{30}\right)^2 = 0.1157$$

$$P(n > 5) = 1 - \sum_{n=0}^5 (1 - \rho)\rho^n = \rho^6 = 0.335$$

(The slide adds six rounded terms and gets 0.339; the exact value is $\rho^6 = 0.335$.)

9. Steady State, Multiple Servers and Simulation

Steady state with M servers (slide 23):

- Infinite population with random arrivals or service: utilisation $\rho = \lambda/(M\mu)$ must be **strictly less than 1**.
- $\rho = 1$ is possible only if both arrivals and service are deterministic.
- Finite population: ρ can be any positive value.

Why not 100%? With randomness, a server idle during a random gap can never recover that time, so if average demand equals capacity the backlog drifts upward without limit.

Simulation (slides 24–26): for complex queues, non-exponential distributions or whole networks, analytical formulas do not exist. Simulation runs numerical experiments to answer what-if questions; the simple M/M/1 results are used to **verify** the simulation model. It can be expensive and must be done carefully.

From the textbook (Hicks, Ch. 16): simulation is used when a closed-form ("analytical") solution is not tractable; it imitates the system over time with random numbers drawn from the input distributions.

10. Exam Checklist

- [] LP elements: variables, parameters, objective, constraints, non-negativity; linearity.
- [] Formulate from a word problem: one constraint per resource, $\leq$ for capacities.
- [] Graphical method: intercepts, feasible region, slide the objective line, corner point; compare corners.
- [] Minimisation with $\geq$ constraints: move the cost line toward the origin.
- [] Queue structure (channels $\times$ phases); Poisson arrivals, exponential service.
- [] Convert rates to the same unit ($\mu = 60 \div$ service minutes).
- [] M/M/1 formulas; Little's law; $T_s = T_q + 1/\mu$; $P(n > k) = \rho^{k+1}$.
- [] Steady state requires $\rho < 1$ with randomness.